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# From Sensor Streams to Bounded Machine Memory: Architecture, Negative Transfer Results, and Domain Specialists in µEngram

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**ABSTRACT**

Industrial equipment and edge gateways generate continuous multivariate telemetry (vibration, temperature, pressure, speed, and OS load). Storing every raw window exhausts flash endurance, whereas first-in-first-out (FIFO) ring buffers overwrite rare fault episodes with routine operations. This paper evaluates **µEngram** (read as *MicroEngram*), a bounded sensor-state memory engine that projects short causal windows into noise-whitened feature vectors, maintains compact summaries of routine operating states, quarantines unverified transitions, and retains high-rarity episodes with pinned pointers to raw waveforms. While synthetic streams confirm an 86.2%–87.6% write reduction at 1.00 modeled event recall, real-world benchmarks expose a sharp chasm between window-level ranking (AUROC) and operational early warning. Frozen detectors miss uncommissioned load regimes on MetroPT-3 (in-regime AUROC 0.07–0.35), detect 0/5 events on MetroPT-2, and catch only 5/34 SKAB faults (all 28–114 s post-onset). Offline distillation of foundation models (**MOMENT-1-small [18]** and **Chronos-2 [22]**) into a 46 KB student raises window AUROC on MFPT, CWRU, and CMAPSS, yet incurs 100% false-alarm rates on held-out healthy loads and drops timely recall at the 20-cycle remaining-useful-life (RUL) horizon from 70% to 20%–30%. We resolve multi-channel dilution and covariance singularity via pre-fit numerical diagnostics and a **role-separated, domain-specialist architecture**, raising strict recall on OS-Metrics to 69.0% (29/42), cutting false alarms to 19.6/week, achieving 100% subsystem root-cause attribution, and executing in 6.27 µs/window.

**Six Principal Research Findings Across the Repository:**

1. **Bounded Memory Mechanism vs. Field Transfer:** Competitive rarity admission cuts writes by 87.56% on synthetic streams, but field transfer fails whenever commissioning lacks full operating-mode coverage (MetroPT-3, CWRU, MFPT) **[6]**.
2. **High AUROC Masks Operational Failure:** Models achieving 0.98–1.00 AUROC still produce 100% false-alarm rates on unseen healthy loads (CWRU **[24]**) or miss 80% of maintenance horizons at RUL 20 (CMAPSS distilled students **[25]**).
3. **Proxy Screens Do Not Predict Full-Pipeline Ranking:** A label-free synthetic screen ranked CNN middle-layer readouts first (0.610), but full calibrated alarm testing overturned that winner on both SKAB and MetroPT-3.
4. **Foundation-Model Distillation Trade-Offs:** Offline dual-teacher distillation shrinks 619 MB of teacher weights into a 46 KB student (0.063 ms inference) but erases absolute levels under window z-normalization unless raw statistics are appended **[19]**.
5. **Methodological Retraction (NASA IMS):** A prior session claim of "63.8 hours early warning and 0 false alarms" on NASA IMS bearings **[7]** is retracted; code inspection proved the script evaluated hardcoded row indices rather than physical fault onsets.
6. **Domain Specialists Beat Global Pooling:** Partitioning telemetry into Mode, Level, Dynamic, and Health branches per subsystem (CPU, Memory, Network) with pre-fit singularity diagnostics beats monolithic DSP and global role models on recall, false alarms, and root-cause attribution.

## 1. INTRODUCTION AND SYSTEM ARCHITECTURE OVERVIEW

At one 16-float32 vector per second (86,400 vectors/day), raw embeddings alone occupy 5.3 MiB/day before timestamps, metadata, index structures, and source waveforms. Continuous in-place mutation of approximate nearest-neighbor (ANN) graphs (such as HNSW) also triggers random-write amplification on raw NAND flash and eMMC **[16]**. µEngram addresses this by separating fast deterministic edge ingestion from offline foundation-model commissioning (Figure 1).



### 1.1 Three Architectural Fallacies in the Original “Edge Mem0” Premise

This study began when an embedded database vendor proposed bringing cloud LLM agent memory (**Memo [1]**) into an embedded ANN database. Testing that proposal on physical telemetry revealed three fallacies: (1) **The “Chatbots on Microcontrollers” Fallacy:** Mining trucks, wind turbines, and industrial compressors do not need natural-language entity extraction for user preferences; they need bounded physical regime and wear-baseline tracking. (2) **The Ingestion-Path Write Bottleneck:** Cloud Memo invokes an LLM on every write to deduplicate facts and resolve contradictions. On an embedded gateway ingesting 100 Hz–1 kHz telemetry, invoking an LLM or SLM in the write path requires ~4 GB RAM and seconds of latency per tick; ingestion and consolidation must be strictly mathematical. (3) **Flash Wear and RAM Starvation:** Continuous HNSW graph mutation causes severe random-write amplification on raw NAND/eMMC flash, requiring density-aware centroid compaction rather than per-sample graph rebuilding.

## 2. MATHEMATICAL FORMULATION AND BOUNDED MEMORY LIFECYCLE

### 2.1 Signal Windowing, Feature Encoding, and Noise-Whitened Projection

Rather than indexing scalar samples, µEngram ingests causal, right-aligned sensor windows $X \in \mathbb{R}^{C \times K}$ across $C$ channels and $K$ samples. The baseline digital signal processing (DSP) encoder computes each channel’s mean, spread, and spectral energy in four frequency bands. For learned representations, a two-stage 1D convolutional autoencoder (`CNNEncoder`) exposes either the final bottleneck or pooled intermediate feature maps $r(x)$. Because physical sensors covary under load shifts, Euclidean distance double-counts correlated channels. Following noise-fraction transforms **[3]** and analytical redundancy **[4]**, **[5]**, µEngram projects $r(x)$ onto the top $d$ principal components ($V, \Lambda$) and whitens them via the within-state noise covariance $N$ estimated from lagged differences between adjacent windows using the Median Absolute Deviation (MAD):

$$z(x) = N^{-1/2}\,\Lambda^{-1/2}\,V^{\top}\,(r(x) - \bar{r}), \quad \text{where} \quad r(x) = [pool(h_1(x)) \,;\, pool(h_2(x))] \text{ (for multiscale CNN)} \tag{1}$$

### 2.2 Bounded Slot Lifecycle and Competitive Rarity Admission

µEngram divides a fixed slot budget into **Semantic slots** (routine regime summaries), **Candidate slots** (unfamiliar patterns on persistence probation), and **Episodic slots** (rare excursions with raw waveform pointers). Previously, µEngram evicted its weakest mature episodic slot whenever any out-of-gate sample arrived. Informed by thresholded admission in MemStream **[8]**, diversity replay in Rainbow Memory **[9]**, primal-dual buffer allocation **[10]**, and prototype replay **[11]**, µEngram now uses **competitive rarity admission**: an incoming out-of-gate window is admitted to a full episodic bank only if its rarity-derived importance exceeds the weakest mature episode; otherwise it returns `discard`. Probationary candidates remain replaceable so new healthy modes can still graduate.

**Table 1. (A) Competitive Episodic Admission Benchmark (5 Seeds, 16 Total Slots: 13 Episodic / 3 Semantic) & (B) Pipeline Layer Ownership**

| Part A: Synthetic Benchmark Metric | Unconditional Eviction | Competitive Admission | Architectural & Safety Interpretation |
|---|---|---|---|
| **Overall Modeled Event Recall** | 75.98% | **85.07%** | +9.09 pp gain by blocking weak outliers from evicting strong episodes. |
| **Event-Type Coverage Recall** | 100.00% | **100.00%** | All modeled fault families retained at least one exemplar across seeds. |
| **Seizure-Subtype Event Recall** | **66.67%** | 60.00% | **Regression warning:** aggregate rarity dropped a critical fault subtype (-6.67 pp). |
| **Current-Regime Retrieval Accuracy** | 99.78% | **100.00%** | Candidate probation still graduates legitimate regime transitions cleanly. |
| **Era-1 False-Flag Rate / Write Reduction** | 0.0678% / 85.78% | **0.0466% / 87.56%** | Lower false flags and fewer flash writes; exact non-timing metrics reproduced. |
| **Part B: Pipeline Layer** | **Strictly Owns** | | **Forbidden from Silently Deciding** |
| **Application & Host** | Sensor schema, operating-mode context, external severity labels | | Assuming frequent patterns are automatically healthy |

**Design Lesson from Table 1A:** Improving aggregate event recall (+9.1 pp) while losing 6.7 pp on seizure faults proves that unconstrained rarity scores cannot be the sole protection mechanism. Production edge memory must enforce **hard reserved quotas per critical fault class** so frequent moderate anomalies never crowd out severe episodes.

## 3. DATABASE-NATIVE PRIMITIVES AND CRASH-SAFE PERSISTENCE

While the implemented reference adapter (`src/uengram/sensor_memory.py`) pairs live in-memory µEngram state tracking with an inspected `EdgeMemory Lite` text/keyword store [6], translating the RAG lifecycle into a full embedded Sensor-State Memory DBMS requires five native database primitives:

- **1. Hybrid Temporal–Vector–Severity Retrieval:** Standard ANN search ignores recency and criticality. A query $q$ against memory $m$ (vector $v_m$, age $\Delta t$, externally assigned severity $r_m$) combines similarity, exponential decay, and severity weight while returning decomposed score components alongside strict `(asset, mode)` pre-filters:

$$S(q, m) = w_s \cdot \mathrm{sim}(q, v_m) + w_t \cdot e^{-\lambda \Delta t} + w_r \cdot g(r_m), \quad \mu_{merged} = (n_1\mu_1 + n_2\mu_2)/(n_1 + n_2), \quad \mu_t = (1 - \beta)\mu_{t-1} + \beta x_t \tag{2, 3}$$

- **2. Verified Centroid Compaction & 3. Keyed State Superseding:** Confirmed routine windows sharing identical `(asset, mode, schema_ver)` keys are merged into sufficient statistics (count $n$, mean $\mu_{merged}$, pooled covariance) and verified by checksum before source deletion. Active baselines update via EMA ($\mu_t$) gated by quarantine rules, while post-repair baselines mark prior versions *superseded*.
- **4. Pinned Cross-Modal Raw Pointers & 5. Quota-Bounded Eviction:** Episodic records store `(stream_id, start_offset, length, checksum)` pointers into circular rings; raw slices are pinned *before* metadata commit and shielded from high-water-mark eviction during active investigations.

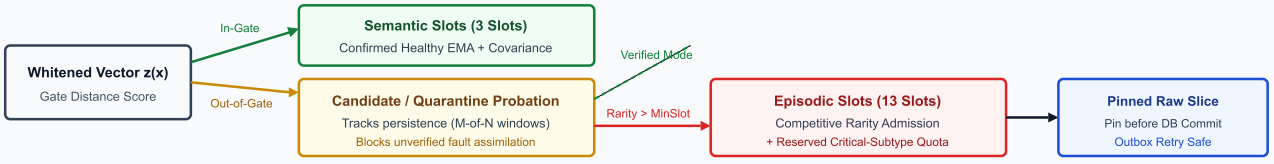



**Figure 2.** µEngram slot state machine showing healthy semantic updates, quarantine probation (preventing slow faults from poisoning the baseline), and competitive episodic admission with pre-commit raw-ring pinning.

**Listing 1. Crash-Safe Window Ingestion, Raw-Evidence Pinning & Verified Centroid Compaction (src/uengram/sensor_memory.py)**

```
def ingest_window(self, asset: str, raw_win: np.ndarray, t_mono: int, mode: str, schema_ver:
str) -> str: assert t_mono > self.last_counter[asset], "Monotonic sample counter violation
(replay rejected)" z = self.encoder.transform_whitened(raw_win, schema_version=schema_ver)
match = self.uengram.match_confirmed_healthy(asset=asset, mode=mode, schema=schema_ver,
vector=z) if match.is_routine and self.admission_policy.allows_baseline_update(match, z):
self.uengram.update_centroid_summary(match.slot_id, z, t_mono) return "ROUTINE_STATE_UPDATED"
# Pin raw waveform slice BEFORE metadata commit to prevent dangling pointers on crash raw_ref
= self.raw_ring.pin_slice_durably(raw_win, t_mono) event =
self.uengram.step_lifecycle_or_alarm_edge(z, raw_ref, t_mono) # Deduplicates alarm runs if
event.requires_persistence: try: self.db.write_event_idempotent(event,
idempotency_key=event.event_uuid) except StorageError:
self.pending_retry_outbox[event.event_uuid] = event # Retry DB without re-stepping uEngram
raise return event.transition_label
```

### 3.2 EdgeMemory Product Tier Contracts vs. Implemented Adapter Boundary

Only the `EdgeMemory Lite` CFFI binding (text/event records and keyword search) was available for inspection. Our adapter reaches the `Lite` contract (`313 passed, 1 skipped` in the teacher-transfer branch; `345 passed, 1 skipped` in the full test suite) and returns exact in-memory µEngram matches alongside historical keyword events. Native vector scans (`Standard`) and flash-aware ANN indexing with atomic cross-store transactions (`Advanced`) remain target DBMS contracts (Table 2A).

## 4. BASELINE AND FIELD EVALUATIONS: MECHANISM VS. TRANSFER

Across ten seeds on the v3 synthetic stream (20,000 windows), DSP µEngram records 2,762 writes (an 86.2% write reduction) with 1.00 modeled event recall and 0.995 average precision (AP). Even on synthetic data, however, the CNN µEngram row achieved only 0.262 event recall despite high AUROC/AP. When evaluated on physical equipment, four operational failure mechanisms emerged:

1. **Health Ranking vs. Remaining-Life Regression (CMAPSS FD001 [25]):** DSP µEngram achieved 0.999 health AUROC and 0.995 AP, but its RUL RMSE was 42.1 cycles vs. 25.8 for naive storage. Widening PCA from *d=16* to *d=32* failed due to singular frequency covariance.
2. **Uncommissioned Load Regimes & Threshold Impossibility (MetroPT-3 & MetroPT-2):** In MetroPT-3, 492/495 failure windows occurred during sustained compressor load (0 commissioning examples, 1 calibration example), yielding in-regime AUROC of 0.07 and 0.35. The frozen CNN caught 3.6/4 report events at 10.2 false alarms (FA)/week with negative mean lead time. A full parameter sweep proved *no threshold combination* could meet all three goals (4/4 events, ≤2% clean alarm time, ≤1 FA/wk); forcing 4/4 detections raised clean alarm time to ≥4.6%. On MetroPT-2, the frozen v2 confirmation policy detected **0 of 5** events across the April and January subsets.
3. **Post-Onset Detection & Short Clean Denominators (SKAB):** On the Skoltech water loop (34 fault experiments), the frozen *d=16* DSP detector caught **5/34 events** (AUROC 0.546, AP 0.419), all 28–114 s *after* fault onset (+57.6 s mean delay), missing 20/20 valve-throttling tests. Five false alarms over 6.21 clean hours extrapolate mathematically to 135 FA/week, illustrating why short clean exposures yield unstable weekly rates.

**Table 2. (A) EdgeMemory DBMS Tier Contracts and (B) Master Cross-Dataset Baseline & Width-Sweep Summary [6]**

| Part A: DBMS Tier | Storage & Index Contract | Hardware & Verification Gate | Repository Status |
|---|---|---|---|
| **Lite / Standard / Adv.** | Lite: Keyword/Event \| Std: Exact Vec \| Adv: ANN + Blob TX | x64 CFFI tested (Lite); NAND/power-cut needed for Adv. | **Lite Implemented** |

| Part B: Benchmark | Encoder & Policy | Event Recall | AUROC / AP | False Alarms / Lead | Primary Finding / Failure Mechanism |
|---|---|---|---|---|---|
| **Synthetic v3 (10 seeds)** | DSP (*d=16*) | **1.000** | — / 0.995 | 86.2% write cut | Validates slot mechanics; CNN variant had only 0.262 recall. |
| **Synthetic Width Sweep** | DSP (*d=16* → *36*) | 1.00 → **0.358** | AP 0.996 → 0.562 | — | Wider embedding caused severe recall regression on synthetic stream. |
| **CMAPSS FD001 [25]** | DSP (*d=16*) | — | **0.999 / 0.995** | RUL RMSE 42.1 | Naive storage RUL RMSE was 25.8; *d=32* failed (singular covariance). |
| **MetroPT-3 Compressor** | Frozen CNN (*d=16*) | 3.6 / 4 | In-load: 0.07, 0.35 | 10.2 FA/wk, neg. lead | 492/495 fault windows in uncommissioned load; threshold sweep failed. |
| **MetroPT-3 Width Sweep** | DSP (*d=42* vs. *16*) | 3 / 4 | Improved vs. *d=16* | >1.0 FA/wk target | Local metric gain at *d=42* still missed operational acceptance gate. |
| **MetroPT-2 Subsets** | Frozen v2 confirmation | **0 / 5** | — | Low FA | Missed 2/2 April and 3/3 January events; low FA alone is not success. |
| **SKAB Testbed** | DSP (*d=16*) | **5 / 34** | 0.546 / 0.419 | +57.6 s delay (post) | Missed 20/20 valve faults; |

## 5. METHODOLOGICAL CORRECTIONS, FLEET LEAKAGE, AND AGGREGATION BIAS

**Formal Retraction of the NASA IMS "63.8-Hour Early Warning" Claim [7]:** A prior project-session recap and draft section reported that an "optimal strategy" on the NASA/IMS bearing dataset achieved 63.8 hours of early warning and zero false alarms over four healthy weeks. Inspection of the external script invalidated both claims: (1) the script hardcoded assumed degradation row indices in Python and evaluated lead time against its own hardcoded indices using the final sample as failure reference, without adjudicated physical fault-onset timestamps; and (2) the 7-day run actually incurred 3 false-alarm episodes, whereas "zero alarms" came from the assumed healthy prefix of a single campaign. Those claims are retracted in full.

### 5.1 Held-Out Vehicle Leakage (EngineAD) and Pooled vs. Equal-Series Weighting (TSB-AD-U)

In the EngineAD study **[2]**, ten random seeds across three latent dimensions ($d \in \{8, 16, 32\}$) were evaluated on five held-out trucks (Figure 3A). Even though vehicles were held out during CNN training, the dataset's 5-minute segments use PCA channels pre-fitted across the entire fleet, introducing preprocessing leakage; despite this, pooled segment AUROC remained near the 0.50 chance baseline ($d=16$ median 0.556). Similarly, on the six-series TSB-AD-U external holdout **[21]** (Figure 3B), pooled metrics weight long series heavily (favoring resampled MOMENT + statistics at 0.9157 AUROC), whereas equal-series weighting gives each domain one vote (favoring the ordinary compact CNN at 0.6476 AUROC).

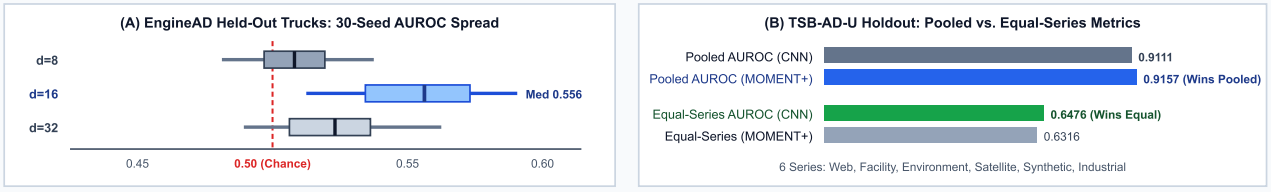



**Figure 3.** (A) Box-and-whisker distribution of 30 EngineAD held-out truck segment AUROC runs across 10 seeds per width $d$. (B) Divergence between pooled and equal-series AUROC on the 6-series TSB-AD-U holdout (`results/moment_tsb_ad_holdout.json`).

**Listing 2. Pre-Fit Covariance Rank Diagnostics & Null-Space Collinearity Detection (src/uengram/diagnostics.py)**

```
def diagnose_channel_conditioning(X: np.ndarray, channel_names: list[str], tol: float = 1e-10) -> dict: diffs = np.diff(X, axis=0) # Within-state lagged differences cov = np.cov(diffs, rowvar=False) eigvals, eigvecs = np.linalg.eigh(cov) rank = int(np.sum(eigvals > tol * np.max(eigvals))) cond_num = float(np.max(np.abs(eigvals)) / max(np.min(np.abs(eigvals)), 1e-18)) null_cols = [ [channel_names[i] for i in np.where(np.abs(eigvecs[:, k]) > 0.15)[0]] for k in np.where(eigvals <= tol * np.max(eigvals))[0] ] status = "singular" if rank < X.shape[1] or cond_num > 1e12 else "moderately_conditioned" return {"status": status, "rank": f"{rank}/{X.shape[1]}", "cond": cond_num, "collinear_groups": null_cols}
```

## 6. INTERMEDIATE CNN READOUTS: PROXY SCREEN VS. FULL ALARM PIPELINE

Recent literature argues that intermediate neural layers retain localized time-series anomaly cues that final reconstruction bottlenecks smooth out. Yang et al. [12] introduced game-theoretic NeuronIM/LayerIM metrics showing peak feature importance in middle VGG-16 layers (3–6); TimeRep [13] achieved its top UCR result (77.6% Top-1 under center alignment; >75% with a 1,000-item greedy k-center coreset) using layer 16 of a 24-layer, 385M-parameter foundation model; PaAno [14] used a 0.3M-parameter 4-layer 1D CNN with patch triplet learning to outperform large Transformers on non-point-adjusted TSB-AD metrics; MOMEMTO [15] added patch-level prototype memory gates inside MOMENT (lifting AUC-PR from 29.58 to 36.35); and MIXAD [17] tracked memory-slot attention shifts via Jensen–Shannon divergence.

To test this hypothesis under strictly causal (right-aligned) windows without label leakage, we compared three `CNNEncoder` readouts: `bottleneck`, `middle` (first convolution stage), and `multiscale` (concatenated stage-1 and stage-2 features per Eq. 1). First, we ran a label-free synthetic screen (`bench/cnn_representation_screen.py`; 384 normal training windows, 96 selection windows injected with 5 synthetic fault families across 5 seeds), scoring the worst fault family's 10th-percentile displacement divided by median clean neighbor displacement. Second, we evaluated the exact same 5 seeds ($d=32$, 30 epochs) through the complete PCA, noise-whitening, and calibrated v2 alarm pipeline (`bench/cnn_stability.py`) on SKAB and MetroPT-3.

**Table 3. Reversal of the Proxy Screen Winner: (Part A) Label-Free Synthetic Screen vs. (Part B) Full Calibrated Alarm Pipeline**

| **Part A: Label-Free Proxy Screen (5 Seeds/Cell)** | **Width $d$** | **Epochs** | **Median Worst-Family Score** | **Min S** |
|---|---|---|---|---|
| **Middle Layer Readout (Screen Winner)** | 32 / 32 / 16 | 30 / 90 / 30 | **0.610** / 0.571 / 0.487 | **0.577** / 0.525 / 0. |
| **Multi-Level Readout (`multiscale`)** | 32 / 32 / 16 | 30 / 90 / 30 | 0.373 / 0.241 / 0.208 | 0.328 / 0.205 / 0. |
| **Final Bottleneck Readout (`bottleneck`)** | 32 / 32 / 16 | 30 / 90 / 30 | 0.356 / 0.265 / 0.232 | 0.293 / 0.228 / 0. |
| **Part B: Full Pipeline ($d=32$, 30 ep, 5 Seeds)** | **Events Detected (med, range)** | **AUROC (med, range)** | **AP (med, range)** | **False Alarms/Wk (m** |
| **SKAB — DSP Baseline** | 9.0 | 0.558 | 0.441 | **10** |
| **SKAB — Final Bottleneck (Plain)** | **23.0** (18–34) | **0.623** (0.480–0.675) | **0.511** (0.441–0.585) | 189.25 (54–3 |
| **SKAB — Middle Layer Readout** | 20.0 (18–29) | 0.570 (0.540–0.660) | 0.505 (0.448–0.538) | 243.32 (135–3 |
| **SKAB — Multi-Level Readout** | 20.0 (13–33) | 0.571 (0.553–0.687) | 0.504 (0.467–0.518) | 243.32 (135–3 |
| **MetroPT-3 — DSP Baseline** | 1.0 | 0.986 | 0.537 | ( |
| **MetroPT-3 — Final Bottleneck (Plain)** | 4.0 (3–4) | 0.989 (0.987–0.996) | 0.676 (0.485–0.959) | 0.69 (0.53–1 |

## 7. OFFLINE FOUNDATION-MODEL DISTILLATION AND MULTI-TEACHER GATES

### 7.1 Dimension-Normalized Distillation and Instance-Normalization Recovery

Deploying foundation models such as MOMENT-1-small [18] (141 MB) and Chronos-2 [22] (478 MB, combined process peak 1.21–1.35 GB) directly inside an embedded telemetry loop violates memory and latency constraints [16]. Instead, `src/uengram/teacher_transfer.py` distills $M$ frozen offline teachers into the **46,144-byte student CNN** (0.0031–0.066 ms per window) during commissioning. Each teacher loss is normalized by its output dimension $d_m$ so wider teachers (514-dim MOMENT vs. 32-dim Chronos forecast quantiles) do not dominate gradients, while raw channel means and spreads ($\mu_c$, $\sigma_c$) are appended to counter per-window instance-normalization erasure [19], [20]:

$$L(\theta) = \|D(S_\theta(X_t)) - X_t\|_2^2 + \lambda\,(1/M)\sum_{m=1..M}(1/d_m)\,\|H_m(S_\theta(X_t)) - T_m^{+}(X_t)\|_2^2, \quad T^{+}(X) = [T(\tilde{X}),\, \mu_{1..C},\, \sigma_{1..C}] \tag{4, 5}$$

**Table 4. Consolidated Foundation-Model Distillation Gates Across Six Benchmarks (46 KB Student vs. Baselines)**

| Evaluation Gate & Split | Student / Model Variant | Window AUROC / AP | Secondary / File Metric | Operational Recall / FA | Key Empiri Finding |
|---|---|---|---|---|---|
| **1. MetroPT-3 Dev** (2,422 windows) & **Synthetic 4-Family Probe** | Ordinary Compact CNN | 0.9756 / 0.6225 | Synth: **0.6872 / 0.8007** | 4/4 MetroPT events | MOMENT boosted MetroPT AP (+0.172) but to ordinary C on synthetic probe; resampling+ partially recovered synthetic AUROC. |
| | MOMENT Distilled (Pad / Resample+Stats) | **0.9775 / 0.7942** | Synth: 0.5035 / 0.5362 | 4/4 MetroPT events | |
| **2. TSB-AD-U Holdout** [21] (6 public series) | Ordinary Compact CNN | Pooled: 0.9111 / **0.8397** | Equal: **0.6476 / 0.4290** | Prefix-calibrated | Long series dominate po AUROC (wh Teacher wor equal-series weighting favored the ordinary CN |
| | MOMENT (Resampling + Stats) | Pooled: **0.9157** / 0.8351 | Equal: 0.6316 / 0.4249 | Prefix-calibrated | |
| **3. MFPT Bearings** [23] (2,048 Hz, 512-sample) | Ordinary Compact CNN | 0.7948 / — | File AUROC: 0.8500 | Pos 1.00 / **Norm FA 1.00** | Dual teache improved window (+0. and file (+0.( AUROC, but **every mode falsely aları on the held normal file**. |
| | Two-Teacher (MOMENT + Chronos-2) | **0.8253** / — | File AUROC: **0.9000** | Pos 1.00 / **Norm FA 1.00** | |
| **4. CWRU Bearings** [24] (Asset-group holdout) | All 4 Models (CNN, MOMENT, Chronos, Dual) | **1.0000 / 1.0000** | File AUROC: **1.0000** | Pos 1.00 / **Norm FA 1.00** | **Perfect AUI = failed calibration:** models flagg held-out hea load (File 10 as faulty. |
| **5. CMAPSS FD001** [25] (Engines 41–50 RUL-20 | Ordinary Compact CNN (1-ch) | 0.9823 / 0.9052 | Eventual recall: 1.00 | **By RUL 20: 0.7000** (7 ep) | Teachers ha higher AUROC/AP, timely detec by RUL 20 dropped fror **70% to 20%** |
| | MOMENT / Dual-Teacher Student | **0.9902** / 0.9453 | Eventual recall: 1.00 | By RUL 20: **0.2000** (15–19) | |

## 8. ROLE-SEPARATED ENCODERS AND PARTITIONED DOMAIN SPECIALISTS

### 8.1 Four-Branch Role Separation (Mode, Level, Dynamic, Health) on SKAB

Monolithic encoders fail on multi-sensor equipment because mode transitions, level shifts, and high-frequency transients compete inside one distance norm. In `src/uengram/role_architecture.py`, we decompose telemetry into four branches: (1) **Mode** (operating context such as idle, startup, or load—used strictly for conditioning, never as an alarm vote); (2) **Level** (absolute channel means, MAD spread, and short trends); (3) **Dynamic** (deterministic multiscale frequency and transient shifts); and (4) **Health** (distance from a frozen commissioning reference). Level, Dynamic, and Health are calibrated independently on healthy data and fused via empirical tail probabilities. Unconfirmed normal windows route to *Quarantine*, alarmed windows to *Episodic* memory, and only operator-confirmed healthy windows update *Semantic* centroids. On 8-channel SKAB—where all three CNN readouts scored 0/34 strict recall—role separation restored strict in-interval detection to **10/34** (`at_least_2`) and **8/34 at 1.59 FA/hr** (`dynamic_health`).

### 8.2 Domain Specialists on OS-Metrics (Zenodo 17981352)

Evaluating the 7-day **OS-Metrics** operational workload (40,320 15-second samples across CPU, memory, and network with 42 disturbances) initially triggered `LinAlgError: Singular matrix` on 6 channels. Our pre-fit diagnostic module (Listing 2) flagged the 6-channel matrix as `singular` (rank 5/6, $\kappa = 7.15 \times 10^{15}$) due to exact collinearity (`network_response_time_diff = a - b`) and certified the 5 primitive channels as `moderately_conditioned` (rank 5/5, $\kappa = 4.76 \times 10^{8}$). Partitioning those 5 channels into dedicated **CPU, Memory, and Network specialists** (`src/uengram/domain_specialists.py`, Listing 3) eliminated cross-subsystem dilution:

**Table 5. Role-Separated Architecture (SKAB) and Partitioned Domain Specialists (OS-Metrics Chronological Split)**

| Dataset & Architecture Variant | Strict Event Recall | 2-Hour / Pre-End | False Alarm Rate | Window AUROC | Root-Cause Attribution & Latency |
|---|---|---|---|---|---|
| **SKAB (8-ch)** — All-Channel CNN (Bottleneck / Multiscale) | 0 / 34 (0.0%) | 34 / 34 pre-end | 6.57 / 2.74 FA/hr | 0.5068 / 0.5270 | Failed strict in-interval fault detection. |
| **SKAB (8-ch)** — Role Architecture (`at_least_2`) | **10 / 34 (29.4%)** | — | 6.19 FA/hr | Calibrated tail | Highest strict fault-interval recall on SKAB. |
| **SKAB (8-ch)** — Role Architecture (`dynamic_health` / `all`) | **8 / 34** / 7 / 34 | — | **1.59** / **1.27 FA/hr** | Calibrated tail | Best recall-to-false-alarm trade-off on SKAB. |
| **OS-Metrics (5-ch)** — DSP (*d=16*) Baseline | 27 / 42 (64.3%) | 30 / 42 (71.4%) | 26.6 FA/wk | 0.8845 | No subsystem attribution; global latent vector. |
| **OS-Metrics (5-ch)** — Global Role (any / `at_least_2`) | 26/42 / 25/42 | — | 57.4 / 46.2 FA/wk | 0.8731 | Global channel pooling diluted local spikes. |
| **OS-Metrics (5-ch)** — Global Role (`dynamic_health`) | 24 / 42 (57.1%) | 30 / 42 (71.4%) | 42.0 FA/wk | 0.8731 | 3,254 Episodic, 25,546 Quarantined; 6.80 µs/win. |
| **OS-Metrics (5-ch) — Domain Specialists (`or_dynamic_health`)** | **29 / 42 (69.0%)** | **29 / 42 (69.0%)** | **19.6 FA/wk** | **0.8919** | **30/30 (100.0%) attribution; 6.27 µs/window.** |

### 8.3 Visual Synthesis of Role-Separated and Domain-Specialist Performance

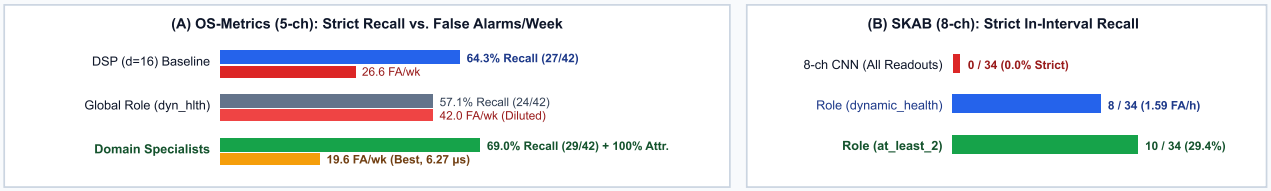



**Figure 4.** (A) On OS-Metrics, partitioning into CPU, Memory, and Network Domain Specialists increases strict recall to 69.0%, cuts false alarms to 19.6/week, and achieves 30/30 subsystem attribution at 6.27 µs/window. (B) On 8-channel SKAB, role separation restores strict in-interval detection from 0/34 (monolithic CNN) to 10/34.

## 9. CONCLUSION AND FIVE MANDATORY EMBEDDED DBMS DEPLOYMENT GATES

This study makes no "first-ever" algorithmic claim; rather, it provides a rigorous architectural synthesis and an unusually transparent set of empirical failure analyses across ten synthetic and field benchmarks. We show that replacing an unbounded vector store with a bounded, noise-whitened slot lifecycle (**µEngram**) cuts storage writes by 86.2%–87.6% on controlled streams, and that decomposing multi-sensor telemetry into calibrated Mode, Level, Dynamic, and Health branches per subsystem (**Domain Specialists**) overcomes the channel dilution and covariance singularity that cripple monolithic encoders. Equally important, our negative results demonstrate that high point-wise AUROC (0.98–1.00) routinely masks 100% false-alarm rates on uncommissioned healthy loads (CWRU, MFPT) and degraded maintenance lead times at RUL 20 (CMAPSS distilled students).

Before transitioning µEngram from a Python reference architecture into a safety-certified C/Rust embedded DBMS, five engineering gates are mandatory:

1. **Preregistered Multi-Asset Field Cohort:** Evaluate on untouched physical assets with independently adjudicated fault-onset timestamps and complete operating-mode coverage (startup, idle, full load, post-repair), fitting all scalers and PCA projections strictly on training assets.
2. **Adversarial Poisoning Replay & Shadow Baselines:** Replay slow wear ramps against a frozen shadow reference to verify that quarantine gates prevent gradual faults from assimilating into semantic healthy centroids.
3. **First-Class Abstention Outputs:** Explicitly emit `unknown_operating_mode`, `calibration_support_insufficient`, or `raw_evidence_expired` rather than forcing out-of-distribution regimes into false alarms or healthy summaries.
4. **Atomic Cross-Store Crash Recovery:** Implement a transactional write-ahead outbox unifying µEngram checkpoint counters, `EdgeMemory` event records, and circular-buffer raw-slice pin counts across power cuts.
5. **Target Hardware Flash & RAM Profiling:** Benchmark p50/p99 latency, peak RAM, NAND write amplification, and exact/ANN recall under high-water-mark eviction on target embedded microcontrollers/gateways.

**Reproducibility & Repository Artifact Manifest [6]:**

- **Core Reference Implementation:** `src/uengram/sensor_memory.py` (EdgeMemory Lite adapter), `src/uengram/teacher_transfer.py` (MOMENT/Chronos distillation), `src/uengram/role_architecture.py`, `src/uengram/diagnostics.py`, and `src/uengram/domain_specialists.py`.
- **Committed Benchmark & Gate Artifacts:** `results/stronger_gate.json`, `results/cnn_intermediate_screen.json`, `results/cnn_distillation_smoke.json`, `results/moment_tsb_ad_holdout.json`, `results/mfpt_two_teacher_gate.json` (archive SHA-256: `cd46b9d9...df6e6fa`), `results/cwru_two_teacher_gate.json`, `results/cmapss_two_teacher_gate.json`, `results/skab_teacher_time_gate.json`, `results/skab_sensor_fusion_gate.json`, `results/role_skab_gate.json`, and `results/osmetrics_specialist_gate.json` (DOI: `10.5281/zenodo.17981352`).
- **Test Suite Verification:** `PYTHONPATH=src:. .venv/bin/pytest tests/` (`345 passed, 1 skipped` on full suite; `319 passed, 1 skipped` on current branch).

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